

Charles Dalton

Gainesville, FL | charles.dalton@ufl.edu | 239-849-3987 | linkedin.com/in/charles-dalton11

EDUCATION

University of Florida

B.S. Nuclear Engineering & B.S. Physics

Aug 2023 – May 2027

GPA: 3.90/4.00

RESEARCH EXPERIENCE

Oak Ridge National Laboratory

May 2026 – Present

Research Intern (DOE SULI) — Radioisotope Science and Technology Division

Oak Ridge, TN

- Characterized double-pulse laser-induced breakdown spectroscopy (DP-LIBS) of flowing argon across a 720-configuration factorial design as a candidate technique for real-time noble-gas monitoring in molten salt reactor off-gas streams.
- Built a Python analysis pipeline linking plasma morphology from ICCD imaging to spectral signal quality, using ANOVA with cluster-robust inference to identify physical predictors of signal-to-noise ratio.

Hartig Lab, University of Florida

Dec 2025 – Present

Undergraduate Research Assistant

Gainesville, FL

- Conduct optical diagnostics (LIBS, emission imaging) of materials in nuclear-relevant environments, including excited-species population analysis and plume vorticity characterization.
- Develop spectral analysis methods including Voigt-profile line fitting with robustness safeguards for low-SNR conditions.

North Carolina State University

May 2025 – Jul 2025

Research Assistant (DOE-Funded REU)

Raleigh, NC

- Designed and built a custom Sample Analysis by Light Transmission (SALT) furnace for UV-Vis and THz-TDS spectroscopy of molten salts.
- Configured the optical system and conducted UV-Vis spectroscopy on molten LiCl-KCl eutectic.

Jung Lab, University of Florida

Jan 2024 – Aug 2025

Undergraduate Research Assistant

Gainesville, FL

- Performed GC-MS experimentation on manufactured plastics and environmental samples to characterize polymer degradation and microplastic composition.

PUBLICATIONS (IN PREPARATION)

Dalton, C., Kitzhaber, Z., Hartig, K., & Andrews, H. “Double-pulse laser-induced breakdown spectroscopy of flowing argon for molten salt reactor off-gas monitoring.” In preparation for *Spectrochimica Acta Part B: Atomic Spectroscopy*.

Borrero, J., Compitiello, A., Realmuto, E., Cook, J., Strack, A., **Dalton, C.**, & Hartig, K. “Spatiotemporal Imaging of Ground-State Al and AlO Reveals Oxygen-Source-Dependent Laser-Ablation Plume Chemistry.” In preparation for *ACS Omega*.

PRESENTATIONS

Dalton, C., Kitzhaber, Z. B., & Andrews, H. B. “Configuration Optimization of Double-Pulse Laser-Induced Breakdown Spectroscopy in Flowing Argon.” *Oak Ridge National Laboratory SULI Summer Poster Session*, Oak Ridge, TN, July 2026. Poster.

Dalton, C. & Bataller, A. “Furnace Design and STRAW Method.” *Advancing Fundamental Molten Salt Modeling Using Ultrafast Spectroscopy, NEUP Annual Review Meeting*, Raleigh, NC, July 2025. Oral presentation.

Sedio, S., Paulson, D., Ritchie, J., **Dalton, C.**, & Jung, S. “Investigation Into the Effects of Location on Microplastic Composition on the University of Florida Campus.” *UF Undergraduate Research Symposium*, April 2024. Poster.

Dalton, C. & Magomo, D. “Smooth Transition Functions in Multiple Dimensions.” *Florida Undergraduate Research Conference*, Miami Gardens, FL, February 2023. Poster.

Dalton, C. & Magomo, D. “Smooth Transition Functions in Multiple Dimensions.” *Mathematical Association of America Conference*, Lakeland, FL, December 2022 (oral).

LEADERSHIP & OUTREACH

Institute of Nuclear Materials Management — Chair

Aug 2026 – Present

- Coordinate with other club officers for planning events and outreach.

American Nuclear Society, UF Chapter — Mentor

Aug 2025 – Present

- Provide sustained academic and research guidance to nuclear engineering mentees; placed mentees into faculty research groups over the academic year.

Society of Physics Students — Outreach Volunteer

Aug 2025 – Present

- Promote physics education through hands-on demonstrations at Alachua County public elementary schools.

HONORS & AWARDS

NEUP Research & Development Award • University Scholars Program • UF Honors University Research Scholars Scholarship • Florida Bright Futures FAS Scholarship

SKILLS & RELEVANT COURSEWORK

Experimental: single- and double-pulse LIBS, ICCD imaging, echelle spectroscopy, UV-Vis, GC-MS, optical alignment, CAD

Analysis & Programming: Python (NumPy, SciPy, pandas), MATLAB, C++, L^AT_EX, Git; Voigt-profile fitting, factorial design, ANOVA, bootstrap inference

Coursework: Quantum Mechanics, Electrodynamics, Statistical Mechanics, Solid State Physics, Thermodynamics, Fluids, Heat Transfer